

Final Presentation: MRI & Image Reconstruction

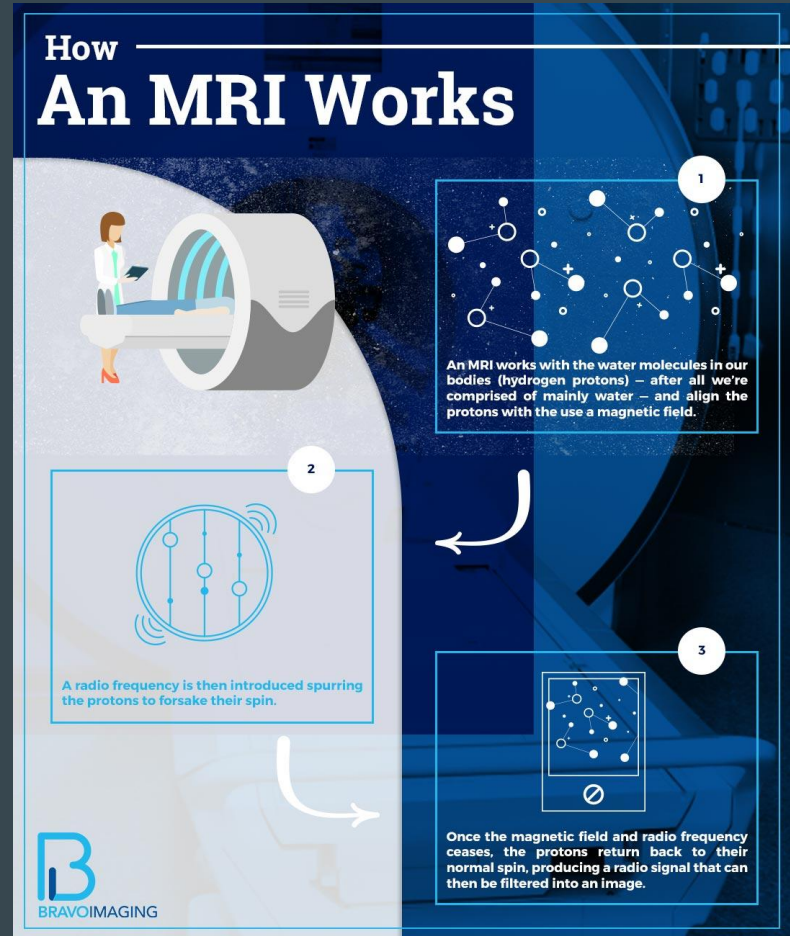
STAT 4830

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Problem Statement & Motivation

A MRI machine has to send out a lot of radio-frequencies, which **takes a long time**. This is **expensive** for hospitals operating MRI machines and **uncomfortable** for patients, who have to stay still while images are taken.

Shorter scan times = more accessible treatment



Technical Approach

$$A = MF$$

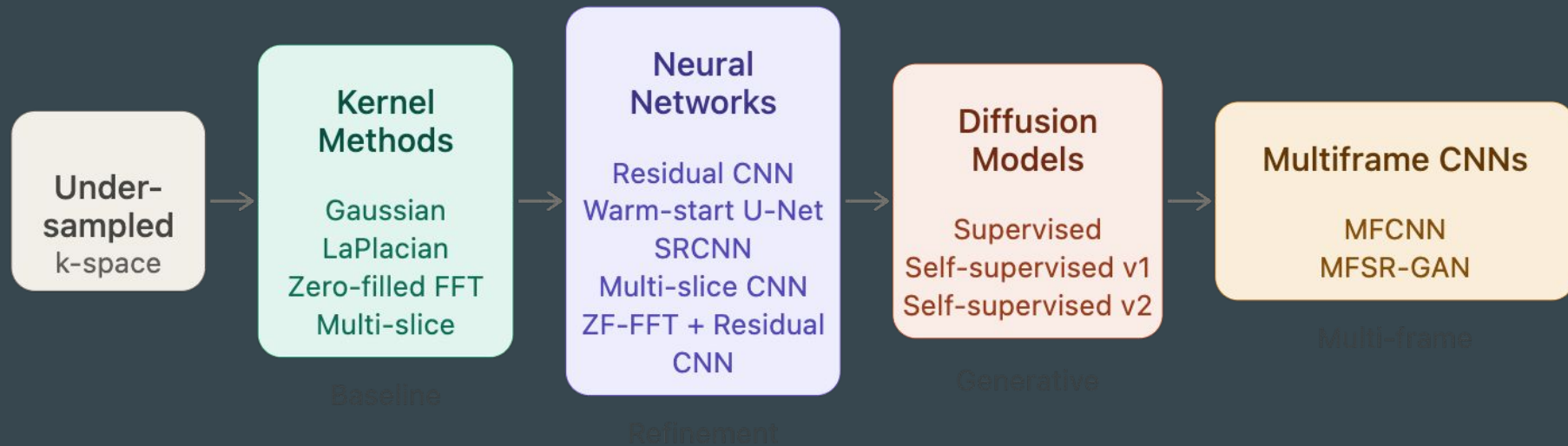
- M = Diagonal Matrix (entries are 1 if frequency is being measured, 0 if not)
- F = Discrete Fourier Transform

Given observed undersampled k-space data y_{obs} , we are looking for α that solves

$$\min_{\alpha \in \mathbb{R}^n} ||A(K\alpha) - y_{obs}||_2^2$$

There are many possible solutions, so we restrict the reconstruction to a class of kernel basis functions for uniqueness. We start with **Gaussian and LaPlacian kernels** .

Research Roadmap

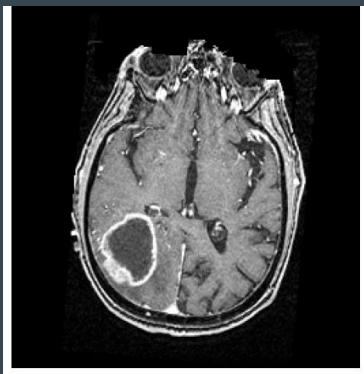


Kernel Reconstructions

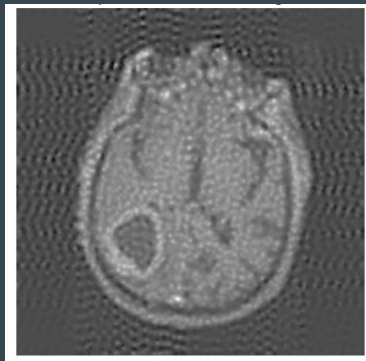
Kernel-Based Reconstruction Results on Single Slice

<i>Metric</i>	<i>LaPlacian Kernels</i>	<i>Gaussian Kernels</i>	<i>Zero-Filled FFT</i>
<i>MSE</i>	0.0186	0.00529	0.010049
<i>PSNR</i>	17.30 dB	22.78 dB	19.98 dB

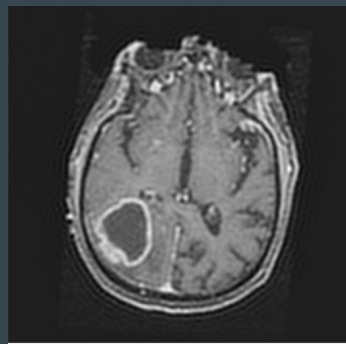
Original Image



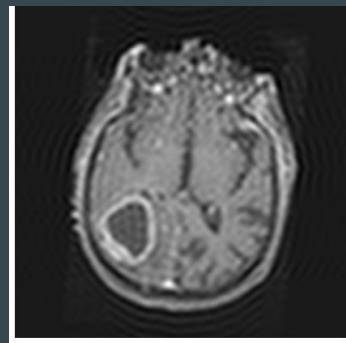
Laplace
Reconstruction



Gaussian
Reconstruction



Zero-filled FFT
Reconstruction



Gaussian kernel reconstruction provides highest clarity.

Multi-Slice Training: Setup

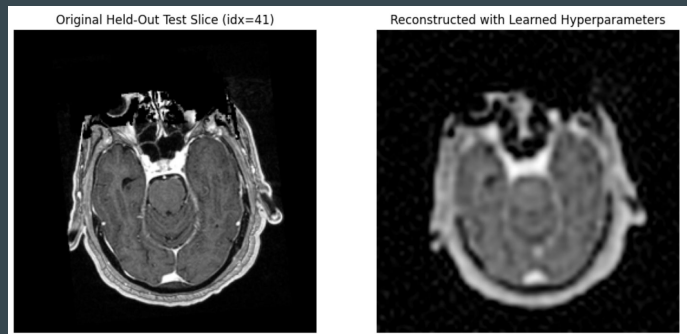
- **Adding a TV penalty:** Our optimization equation largely remained the same, with the addition of a TV regularizer:
- **Parameter Tuning**
 - Training for kernel width, number of kernels, and TV penalty.
 - Using both Gaussian and LaPlacian kernels to align with literature.
- **Training Steps :**
 - Choose candidate hyperparameters,
 - Reconstruct a set of training slices with those hyperparameters
 - Compute reconstruction metrics for each slice.
 - Average metrics across the slices and selected the hyperparameters that performed the best on the test slices.

Multi-Slice Training

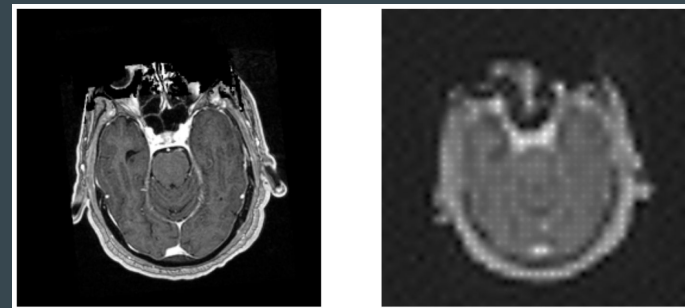
Kernel Type	Sigma	Number of Kernels	TV Weight	Test MSE	Test PSNR
LaPlacian	4.1017	1600	1.3145e-03	0.009615	20.173 dB
Gaussian	2.5	3844	0.0	0.007886	21.032 dB

Like in the single-slice model, the Gaussian kernels performed better as measured by MSE and PSNR.

LaPlacian

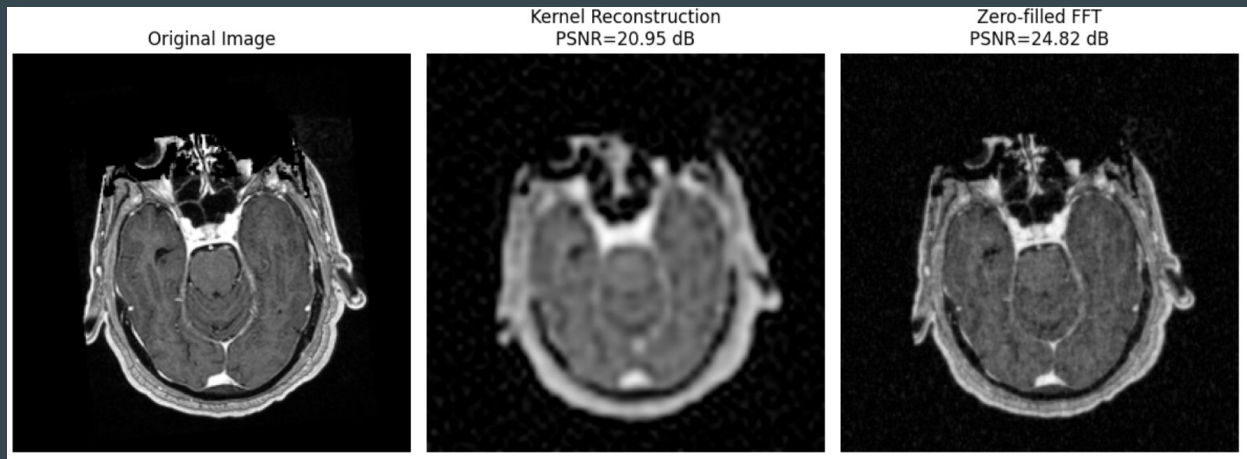


Gaussian



Multi-Slice Training: Zero-Filled FFT

- Lower PSNR for multi-slice zero-filled FFT compared to the single-slice model.
- Simpler method with lower computational time makes zero-filled a good candidate for more advanced models.



Neural Networks

Neural Network Based Reconstruction Setup - Single Slice

- Modern MRI reconstruction often uses learned priors (CNNs/diffusion models)
- Trained a small CNN:
 - 3 layers, 64 channels, 3x3 kernels (standard, small)
 - Input: undersampled MRI slice
 - Predicts a correction: $x^* = x + f(x)$ where
 - x^* is final reconstructed image
 - $f(x)$ is learned correction
 - x is the undersampled input
 - Used L1 loss (common in image restoration tasks): $\|x + f(x) - x_{true}\|$

Neural Network Results - Single Slice

- Modern MRI reconstruction often uses learned priors (CNNs/diffusion models)
- Trained a small residual CNN, then reconstructed the same slice we used the kernel method on



Gaussian kernel still achieves highest PSNR, implying the kernel method is competitive to traditional, learned methods.

New Success Metric: SSIM

$$SSIM(I_{LR}, I_{HR}) = \frac{(2\mu_{I_{LR}}\mu_{I_{HR}} + c_1)(2\sigma_{I_{LR}, I_{HR}} + c_2)}{(\mu_{I_{LR}}^2 + \mu_{I_{HR}}^2 + c_1)(\sigma_{I_{LR}}^2 + \sigma_{I_{HR}}^2 + c_2)}$$

SSIM (structural similarity index measure) provides a quantitative measure of similarity between ground-truth and reconstructed images, measured similarly to how we visually perceive the images. We use it with MSE and PSNR to provide a better holistic view of reconstruction quality.

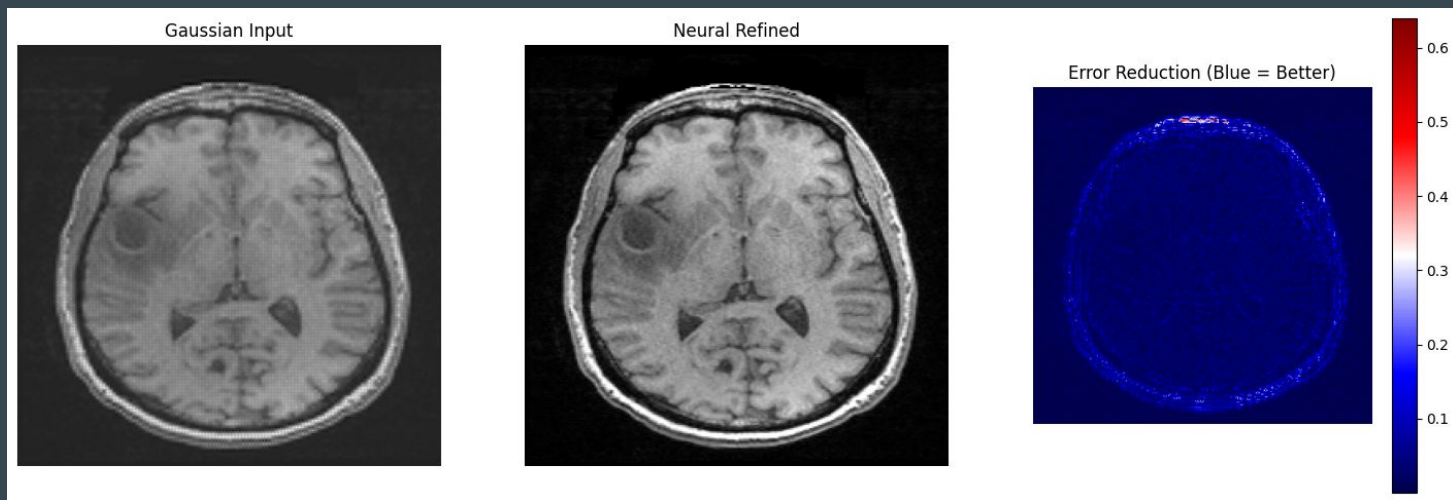
Super Resolution Convolutional Neural Network (SRCNN)

SRCNN is a three-layer CNN for single-image super resolution. Instead of applying kernels, we used SRCNN to reconstruct the entire slice. It made the following improvements:

<i>Metrics</i>	<i>Kernel Optimizer (Gaussian)</i>	<i>SRCNN</i>	<i>Improvement</i>
PSNR	28.78 dB	37.74 dB	8.96 dB
SSIM	0.8786	0.9534	0.0748

Warm-Start Neural Network

Again, used kernel reconstruction as input to a neural network but **only let the neural network fill in the directions the kernel did not already smooth**. We also changed the neural network **architecture to U-Net, rather than ReLU** which is more common in medical applications.



Warm-Start Neural Network Results

<i>Metrics</i>	<i>Gaussian Reconstruction</i>	<i>Warm-Start NN Reconstruction</i>
PSNR	19.581 dB	137.75 dB
MSE	0.011	0
SSIM	0.605	1.0



Achieved a perfect reconstruction of the desired image!

... but we are overfitting to a **single slice**. Can we expand this to a **multi-slice** model?

Residual Convolutional Neural Networks (CNN) - Setup

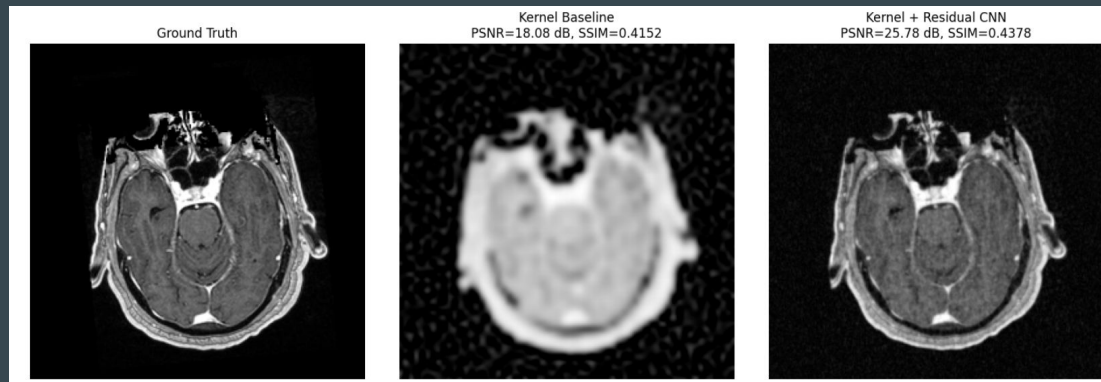
Basic setup: $x_{output} = x_{input} + f(x_{input})$

- E.g. CNN output = Warm-start (from zero-filled FFT) + learned residual correction
- Restriction: Model cannot change measured k-space values
- Tested with zero-filled FFT, Gaussian kernels, and LaPlacian kernels.

	Test MSE	Test PSNR	Test SSIM
Kernel Reconstruction & Residual CNN	0.002479	26.06 dB	0.4153
Zero-filled FFT Reconstruction & Residual CNN	0.000478	33.20 dB	0.6758

Residual CNN - Results

Kernel Reconstructions



Zero-Filled FFT Reconstructions



- Residual CNN consistently shows better performance than kernel reconstruction alone
- Zero-filled FFT outperforms kernel reconstruction on PSNR, MSE, SSIM, and compute time.

Diffusion Models

Diffusion Models

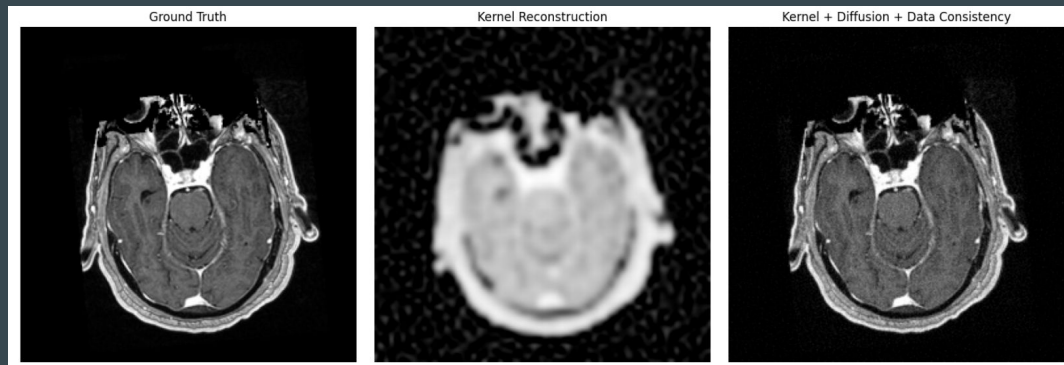
Diffusion models are commonly used in image reconstruction by iteratively adding noise to training images and producing image output from gradually removing noise layers from a pure-noise image to recover the original image.

- Step 1: Train diffusion model on the initial reconstruction (kernel or zero-filled FFT)
- Step 2: Model iteratively corrects the areas of the image that were not measured k-space data (i.e., “real” parts of the image).

	Test MSE	Test PSNR	Test SSIM
Kernel Reconstruction & Diffusion Model Correction	0.000690	31.61 dB	0.7796
Zero-filled FFT Reconstruction & Diffusion Model Correction	0.000772	31.12 dB	0.7620

Diffusion Model: Results

Kernel Diffusion Model



Zero-Filled FFT Diffusion Model



MSE was lower and PSNR and SSIM were higher for the kernel reconstruction compared to the zero-filled FFT, showing that the **combination of the kernel reconstruction and diffusion model actually does better than the baseline approach.**

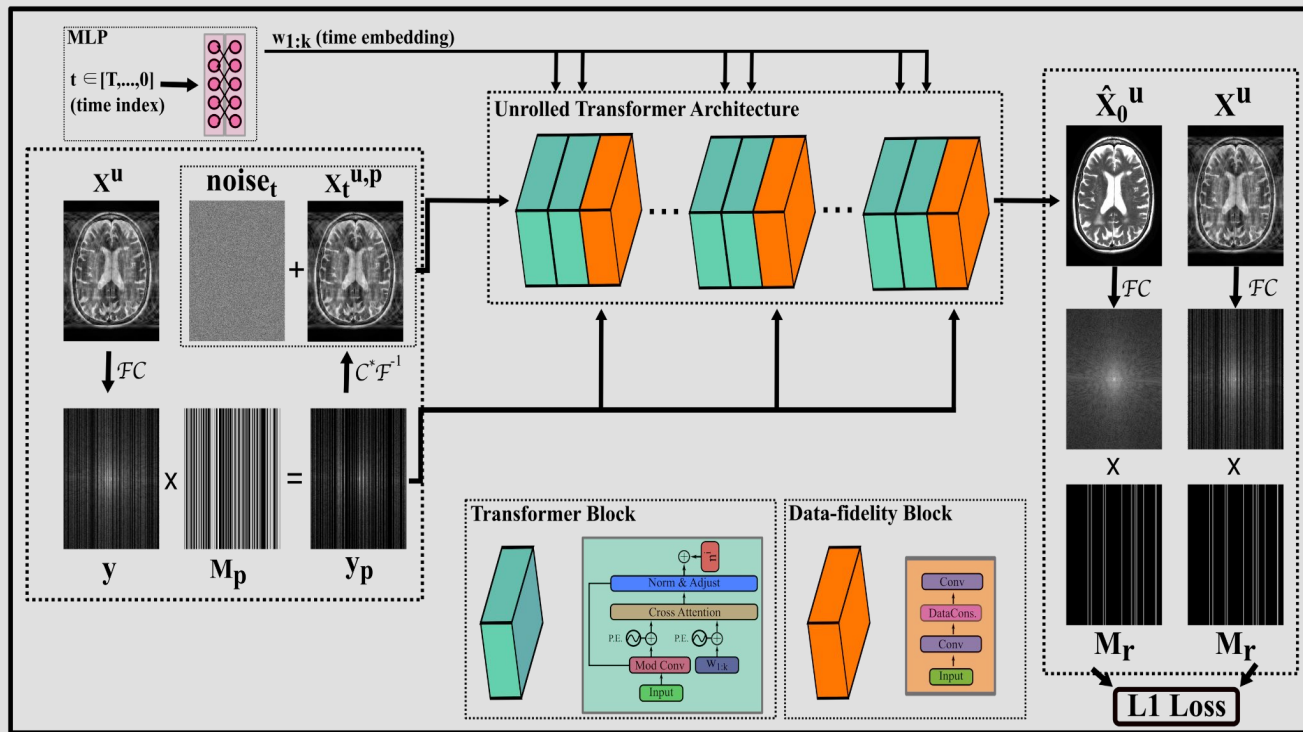
Self-Supervised (Self-Tuning) Diffusion Model

How does the model perform with less available information?

Simulate with a **split k-space masking strategy**.

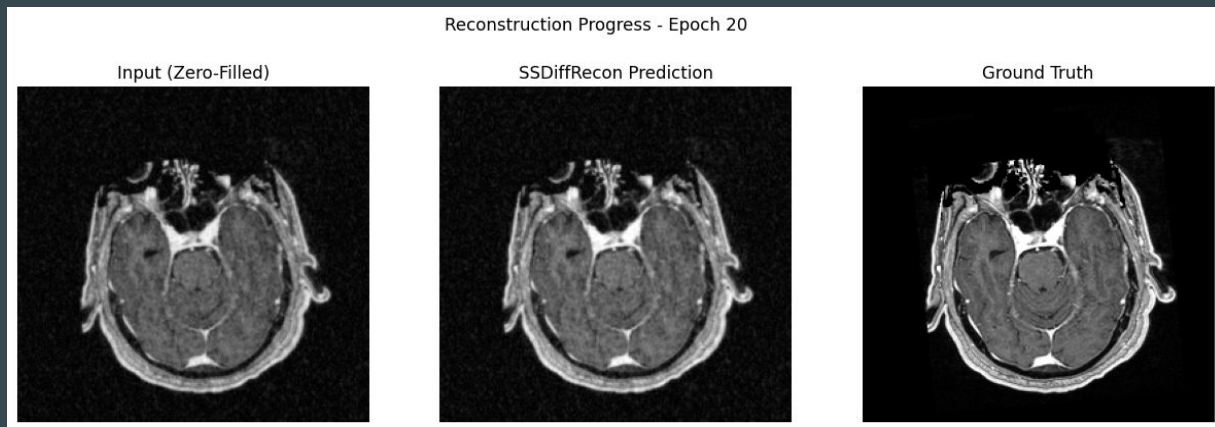
$\mathbf{M}$ = the undersampled mask, $\mathbf{M}_r$ (**train**) = random sample of $\mathbf{M}$ (= 8% here)

$\mathbf{M}_p = \mathbf{M} / \mathbf{M}_r$ (**test**)



Based on paper by Korkmaz et. al (2024)

Self-Supervised Diffusion Model Results



Model is not making changes from zero-filled FFT because it is solving a difficult optimization problem.

Can we change the setup to force the model to make changes?

Loss	MSE	SSIM	PSNR
0.074	0.002	0.427	26.12 dB

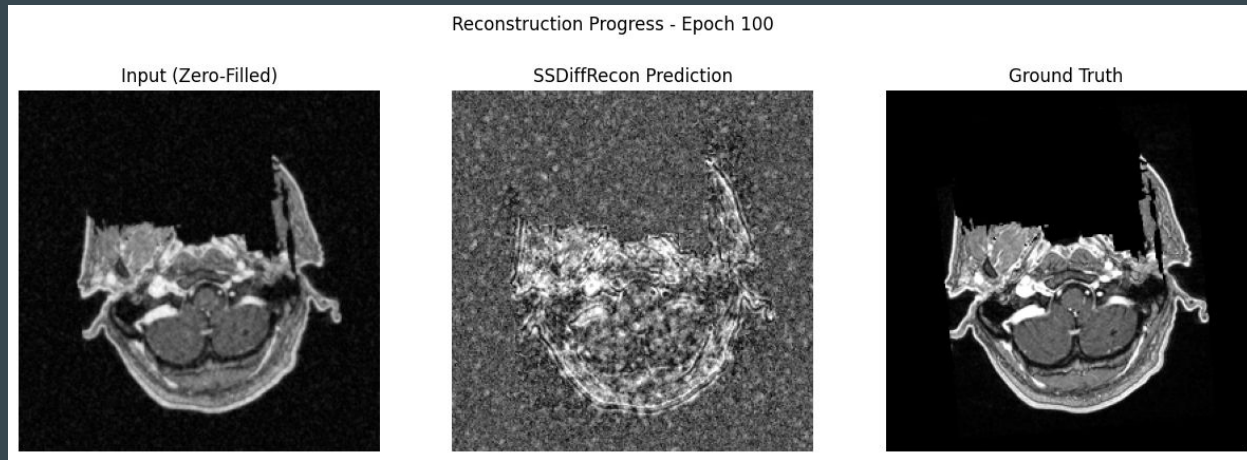
Self Supervised Diffusion Model - attempt 2 (failed)

Step 1: Frequency Weighted Loss: We implemented focal frequency loss because the model had a lot of trouble initially **filling in details** - rather, it duplicated the input with minimal changes -> now the whole screen is black

Step 2: Sigmoidal activation function: Constrains the output to the $[0,1]$ range, preventing extremely bright points in k-space from dominating the image and making model improvements clearer -> better, grainy output but rough shape

Step 3: Cosine Annealing Scheduler with warmup: We start with a **linear scheduler** before switching to cosine annealing so that the model can learn the general shape of the image before dropping the learning rate -> still not great!

Self-Supervised Diffusion Model - version 2 Results



Sigmoidal function really worsened the output!

Did not achieve same results as paper due to smaller training dataset and smaller epoch length.

Loss frequency-weighted	MSE	SSIM	PSNR
0.167	0.110	-0.05	9.59 dB

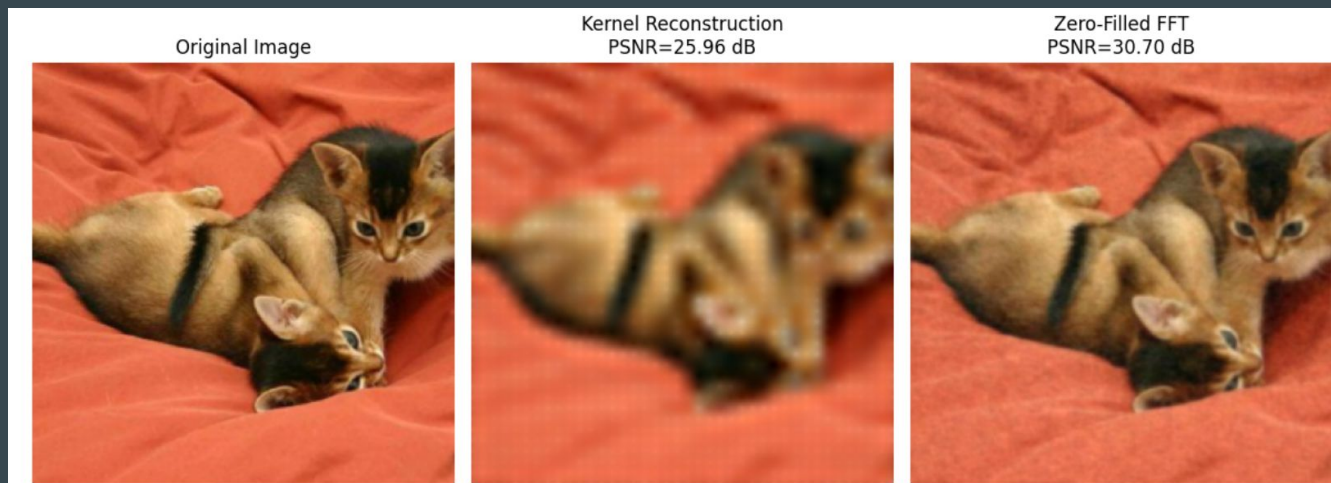
General Images

General Image Hyperparameter Tuning

Kernel Type	Sigma	Number of Kernels	TV Weight	Mean MSE	Mean PSNR
LaPlacian	7.0	1600	1.0e-04	0.004535	24.196 dB
Gaussian	12.0	3600	1.0e-02	0.008821	21.242 dB

- LaPlacian achieved better reconstruction
- Gaussian parameters are much different from the MRI parameters

LaPlacian vs. Zero-filled FFT Reconstruction



Metrics	LaPlacian Reconstruction	Zero-filled FFT Reconstruction
Mean MSE	0.003692	0.002538
Mean PSNR	24.43 dB	26.46 dB

Kernel & Residual CNN Reconstruction



Zero-filled FFT & Residual CNN Reconstruction



Residual CNN Results

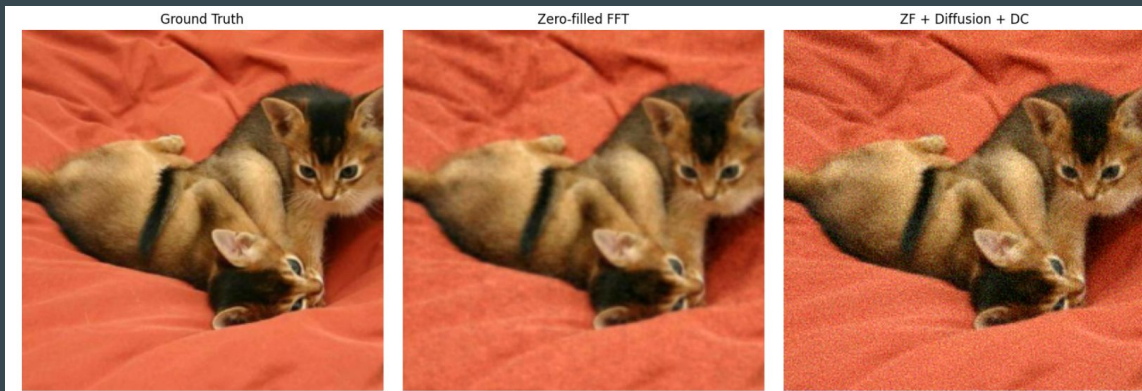
<i>Metrics</i>	<i>Kernel Reconstruction & CNN</i>	<i>Zero-filled FFT Reconstruction & CNN</i>
MSE	0.000833	0.000382
PSNR	30.79 dB	34.18 dB
SSIM	0.8540	0.9104

Zero-filled FFT still achieved better reconstruction, similar to MRI reconstruction results

Kernel & Diffusion Model Reconstruction



Zero-filled FFT & Diffusion Model Reconstruction



Diffusion Model Results

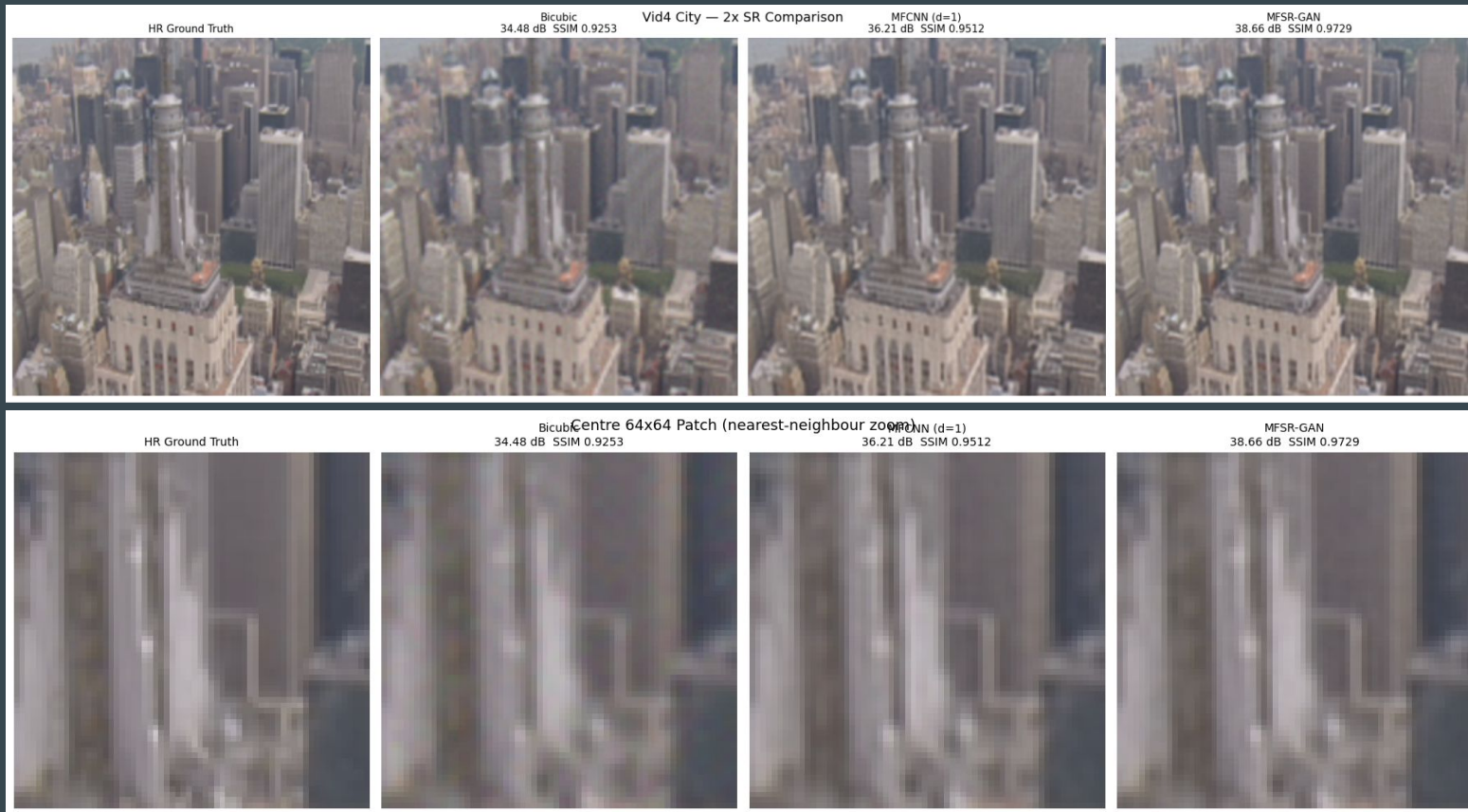
<i>Metrics</i>	<i>Kernel Reconstruction & Diffusion</i>	<i>Zero-filled FFT & Diffusion</i>
MSE	0.004658	0.004504
PSNR	23.32 dB	23.46 dB
SSIM	0.4506	0.4564

Zero-filled FFT still achieved better results, differing from the MRI reconstructions

Both methods achieved worse results than the residual CNN

Multiframe CNN Models

Multiframe CNN Methods



Multiframe CNN Methods

We have implemented two types of Multiframe CNN methods which take into account the context of neighboring frames to help strengthen the reconstruction of each frame

MFCNN (Greaves & Winter, 2016)

A lightweight multi-frame CNN approach. All 5 LR burst frames are bicubic-upsampled to HR size, then concatenated along the channel axis (giving 15 channels for a $d=1$ window). A simple 5-layer conv net maps this to the SR output.

MFSR-GAN (Khan et al., CVPRW 2025)

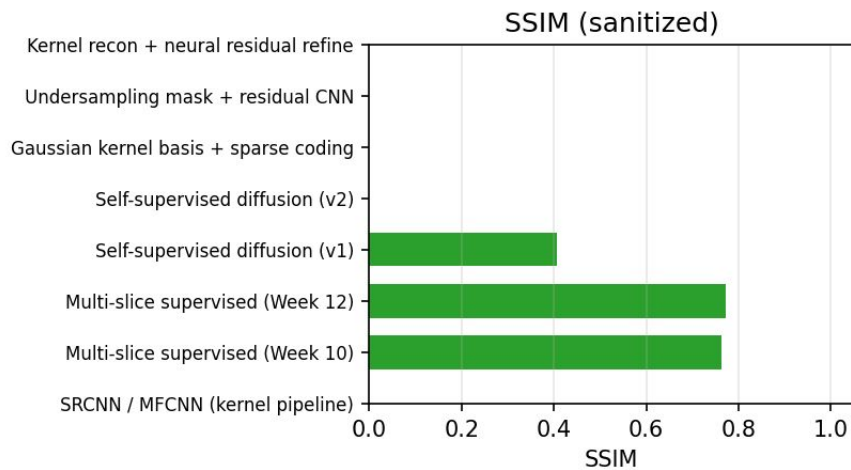
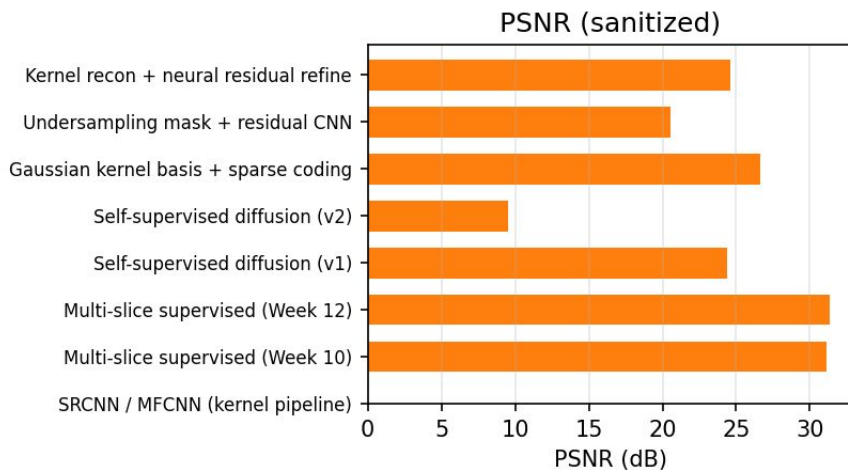
A more sophisticated GAN-based approach with:

- Reference Difference Computation, deformable convolutions for motion-aware alignment, channel attention for multi-frame features, trained with a relativistic GAN loss + L1 loss

Autoresearch and Results

Model Comparison: PSNR and SSIM

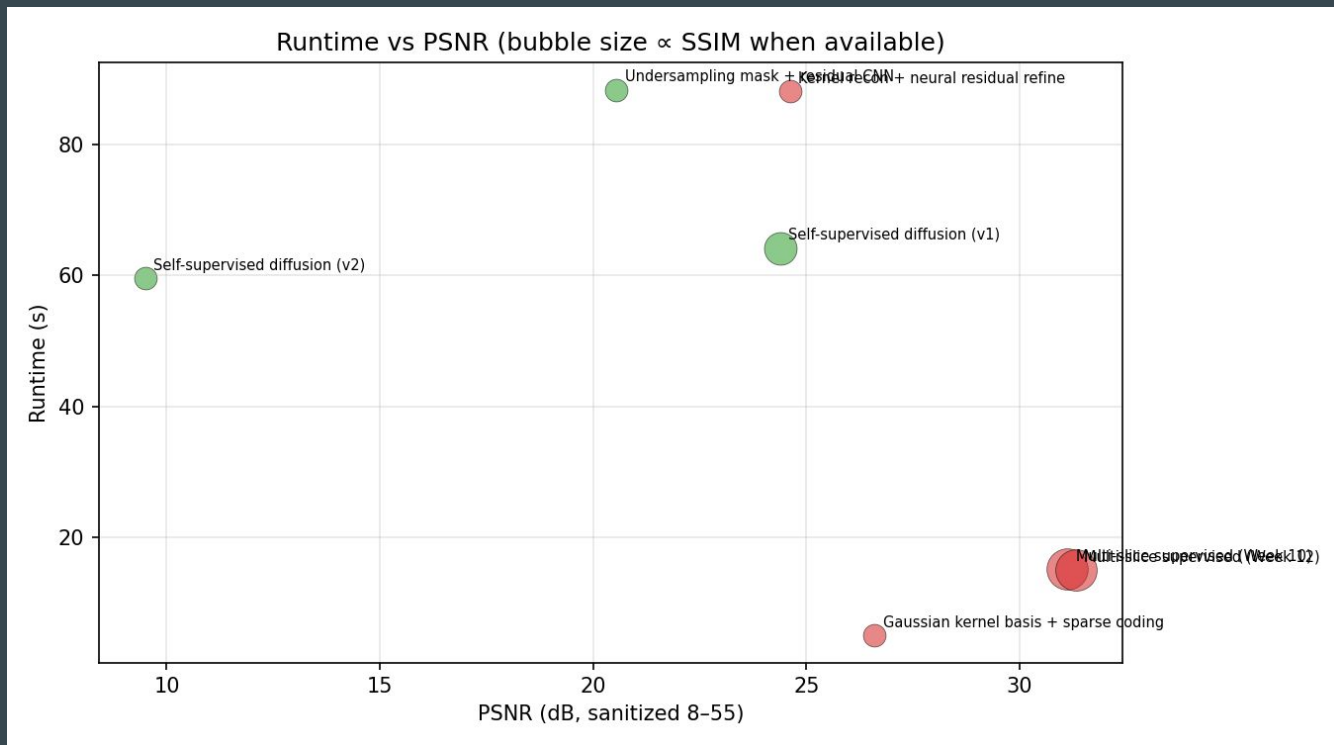
Quality by methodology



What PSNR & SSIM Show

- Multi-slice supervised models lead on both metrics
- SSIM data is missing for several methods, making direct comparison incomplete
- PSNR and SSIM capture different qualities — one number isn't enough

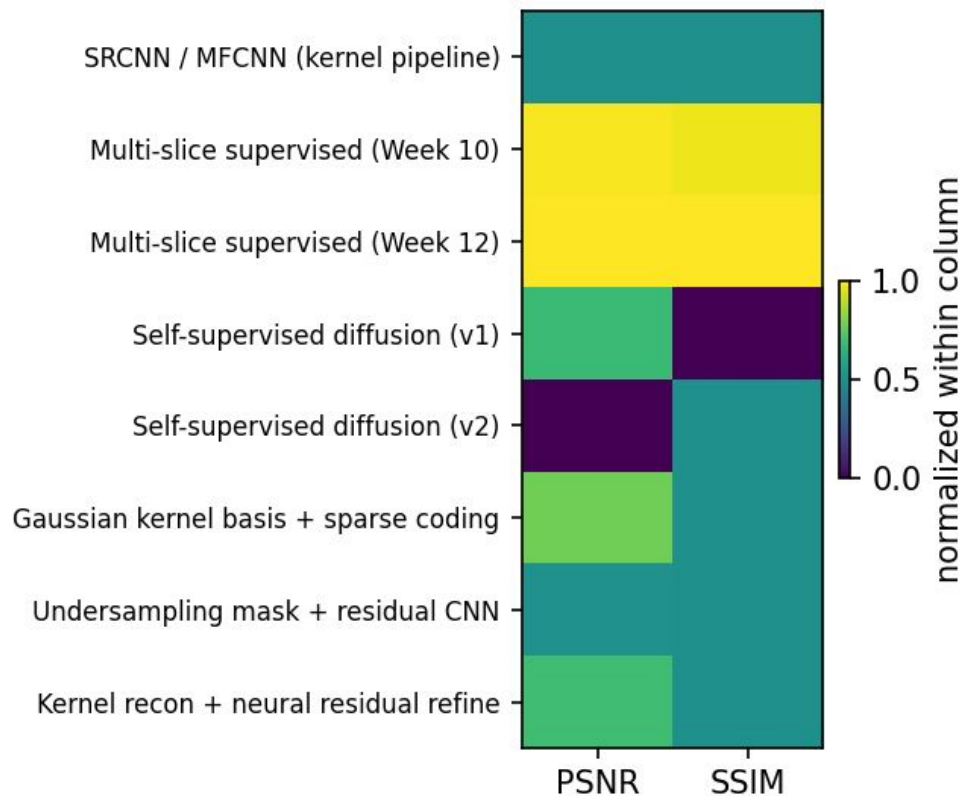
Model Comparison: Runtime vs. Success Metrics



Model Efficiency

- Multi-slice supervised: high PSNR and low runtime (~15s)
- Diffusion models: long runtime (~60-90s) for middling PSNR - costly tradeoff
- Self-supervised diffusion v2: worst of both worlds - slow and lower quality
- For real deployment, multi-slice supervised is the most practical choice
- Diffusion models need significant optimization before clinical use

Quality heatmap (column min-max; PSNR & SSIM only)



No model dominates on both PSNR and SSIM simultaneously

Self-supervised diffusion v1 flips - relatively strong PSNR, weak SSIM

Multi-slice supervised is the most consistent across both metrics

Confirms why using MSE + PSNR + SSIM together matters — each catches something different

Future work: find or design a model that scores well on all metrics, not just one